

HDOC DAY-9 ACTIVITY

C PROGRAMMING ASSIGNMENT REPORT

1. Swapping two numbers using a third variable and without a third variable
2. Sum of the first n natural numbers and sum of their squares
3. Simple interest, compound amount, compound interest and total payable amount

QUESTION 1 – Swapping Two Numbers

Write a menu-driven C program to input two numbers and swap them using a third variable & without using a third variable. Also display the values before and after swapping.

1. Problem Understanding

- **Input:** Two integers.
- **Output:** Values before and after swapping.
- **Menu:** 1 – Swap using a third variable; 2 – Swap without using a third variable; 3 – Exit.
- **Method 1:** Use a temporary variable to exchange the values.
- **Method 2:** Use arithmetic operations to exchange the values without a third variable.

2. Standard Test Cases

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	Choice 1; a = 10, b = 20	Before: 10, 20; After: 20, 10	Before: 10, 20; After: 20, 10	Pass
2	Choice 2; a = 30, b = 12	Before: 30, 12; After: 12, 30	Before: 30, 12; After: 12, 30	Pass
3	Choice 1; a = -5, b = 8	Before: -5, 8; After: 8, -5	Before: -5, 8; After: 8, -5	Pass

3. Algorithm

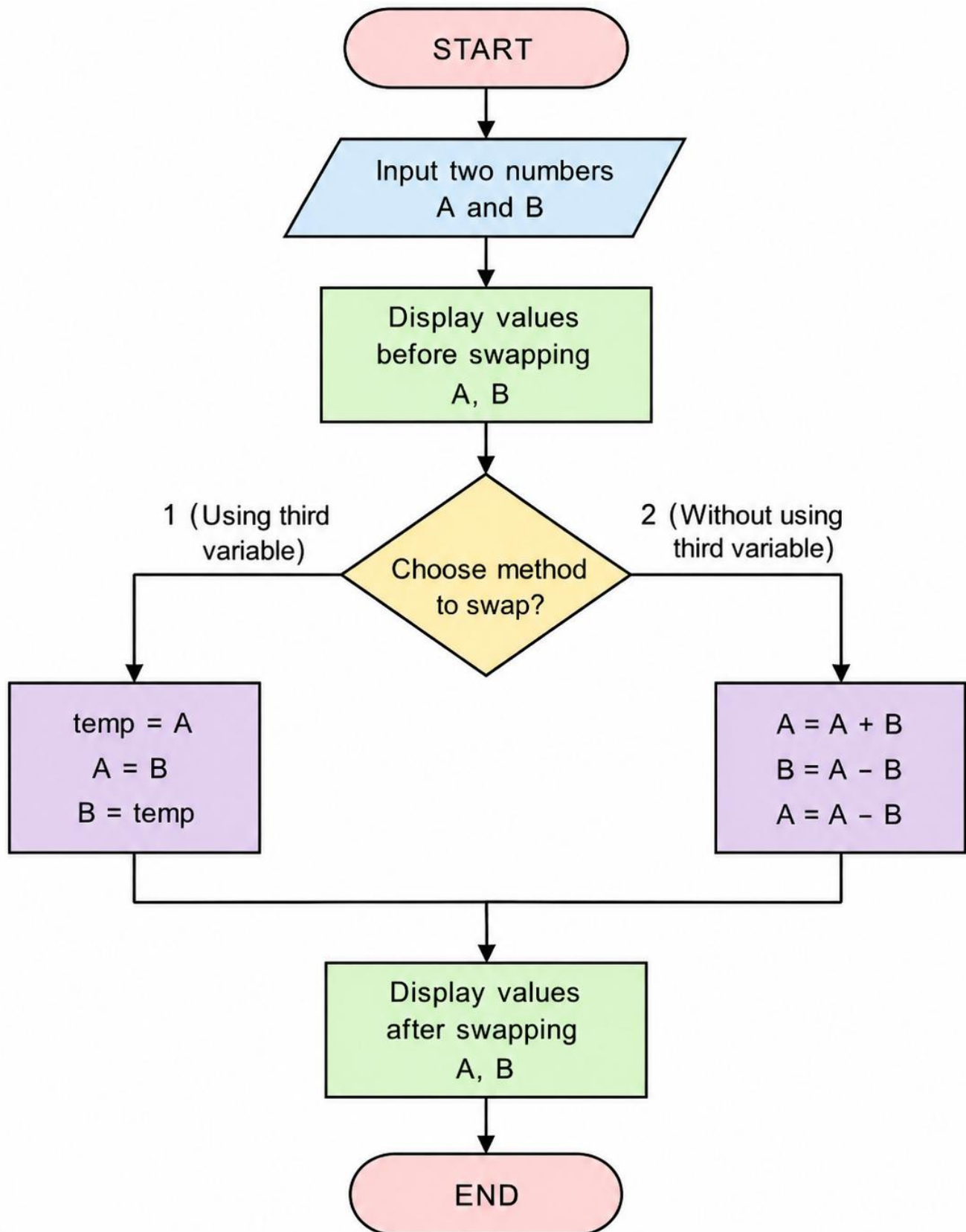
1. Start the program.
2. Declare integer variables choice, a, b, and temp.
3. Display the swap menu and read the user's choice.
4. If choice is 1 or 2, read the two integers and display their values before swapping.

5. If choice is 1, swap using $\text{temp} = a$; $a = b$; $b = \text{temp}$.
6. If choice is 2, swap using $a = a + b$; $b = a - b$; $a = a - b$.
7. Display the values after swapping.
8. If choice is not 1, 2, or 3, display an invalid-choice message.
9. Repeat the menu until the user selects Exit.
10. Stop the program.

4. Pseudocode

1. BEGIN
2. REPEAT
3. Display swap menu.
4. Read CHOICE.
5. IF CHOICE = 1 OR CHOICE = 2 THEN
6. Read A and B.
7. Display A and B before swapping.
8. IF CHOICE = 1 THEN $\text{TEMP} \leftarrow A$; $A \leftarrow B$; $B \leftarrow \text{TEMP}$.
9. ELSE $A \leftarrow A + B$; $B \leftarrow A - B$; $A \leftarrow A - B$.
10. END IF.
11. Display A and B after swapping.
12. ELSE IF CHOICE $\neq$ 3 THEN display Invalid Choice.
13. END IF.
14. UNTIL CHOICE = 3.
15. END

Flowchart – Question 1



6. C Program

```
#include <stdio.h>
int main(void)
{
    int choice, a, b, temp;
    printf("\n--- SWAP MENU ---\n");
    printf("1. Swap using a third variable\n");
    printf("2. Swap without using a third variable\n");
    printf("3. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    if (choice == 1 || choice == 2)
    {
        printf("Enter two integers: ");
        scanf("%d %d", &a, &b);
        printf("Before swapping: a = %d, b = %d\n", a, b);
        if (choice == 1)
        {
            temp = a;
            a = b;
            b = temp;
        }
        else
        {
            a = a + b;
            b = a - b;
            a = a - b;
        }
        printf("After swapping: a = %d, b = %d\n", a, b);
    }
    else if (choice != 3)
    {
        printf("Invalid choice. Please try again.\n");
    }
    return 0;
}
```

7. Verification of All Test Cases

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
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1	Choice 1; a = 10, b = 20	a = 20, b = 10	a = 20, b = 10	Pass
2	Choice 2; a = 30, b = 12	a = 12, b = 30	a = 12, b = 30	Pass
3	Choice 1; a = -5, b = 8	a = 8, b = -5	a = 8, b = -5	Pass

8. Compilation and Execution Evidence

```

kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)-[~/Desktop]
$ ./question1

--- SWAP MENU ---
1. Swap using a third variable
2. Swap without using a third variable
3. Exit
Enter your choice: 1
Enter two integers: 10
20
Before swapping: a = 10, b = 20
After swapping: a = 20, b = 10

(kali@kali)-[~/Desktop]
$ ./question1

--- SWAP MENU ---
1. Swap using a third variable
2. Swap without using a third variable
3. Exit
Enter your choice: 2
Enter two integers: 20
12
Before swapping: a = 20, b = 12
After swapping: a = 12, b = 20

(kali@kali)-[~/Desktop]
$ ./question1

--- SWAP MENU ---
1. Swap using a third variable
2. Swap without using a third variable
3. Exit
Enter your choice: 1
Enter two integers: 5
8
Before swapping: a = 5, b = 8
After swapping: a = 8, b = 5

(kali@kali)-[~/Desktop]
$

```

QUESTION 2 – Sum of Natural Numbers

Write a menu-driven C program to find the sum of the first n natural numbers and sum of squares of the first n natural numbers.

1. Problem Understanding

- **Input:** A positive integer n.
- **Output:** Sum of 1 to n, or sum of squares 1^2 to n^2 , according to the selected menu option.
- **Menu:** 1 – Sum of first n natural numbers; 2 – Sum of squares; 3 – Exit.
- **Validation:** n must be a positive integer.

2. Standard Test Cases

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	Choice 1; n = 5	Sum = 15	Sum = 15	Pass
2	Choice 2; n = 5	Sum of squares = 55	Sum of squares = 55	Pass
3	Choice 1; n = 10	Sum = 55	Sum = 55	Pass

3. Algorithm

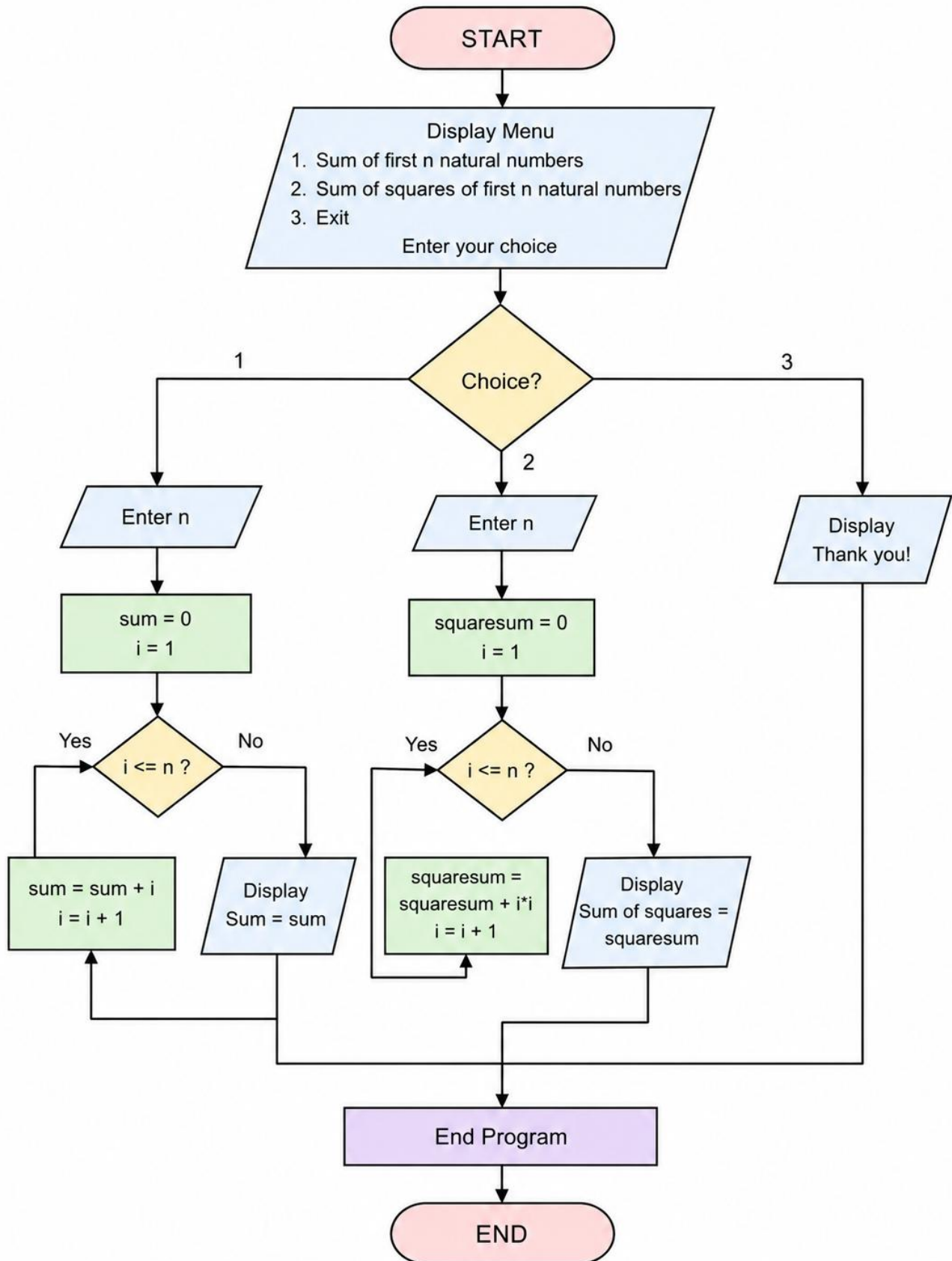
1. Start the program.
2. Declare variables choice, n, i, sum, and sumSquares.
3. Display the menu and read the choice.
4. If choice is 1 or 2, read n.
5. If n is less than 1, display a validation message and return to the menu.
6. If choice is 1, initialize sum = 0 and add each number from 1 to n.
7. If choice is 2, initialize sumSquares = 0 and add $i \times i$ for each i from 1 to n.
8. Display the calculated result.
9. Repeat until Exit is selected.
10. Stop the program.

4. Pseudocode

1. BEGIN
2. REPEAT
3. Display sum menu.
4. Read CHOICE.
5. IF CHOICE = 1 OR CHOICE = 2 THEN
6. Read N.
7. IF $N < 1$ THEN display Invalid input.
8. ELSE initialize the selected sum to 0.
9. FOR $I \leftarrow 1$ TO N
10. IF CHOICE = 1 THEN $SUM \leftarrow SUM + I$.
11. ELSE $SUMSQUARES \leftarrow SUMSQUARES + I \times I$.
12. END FOR.
13. Display the selected sum.

14. END IF.
15. ELSE IF CHOICE $\neq$ 3 THEN display Invalid Choice.
16. END IF.
17. UNTIL CHOICE = 3.
18. END

Flowchart – Question 2



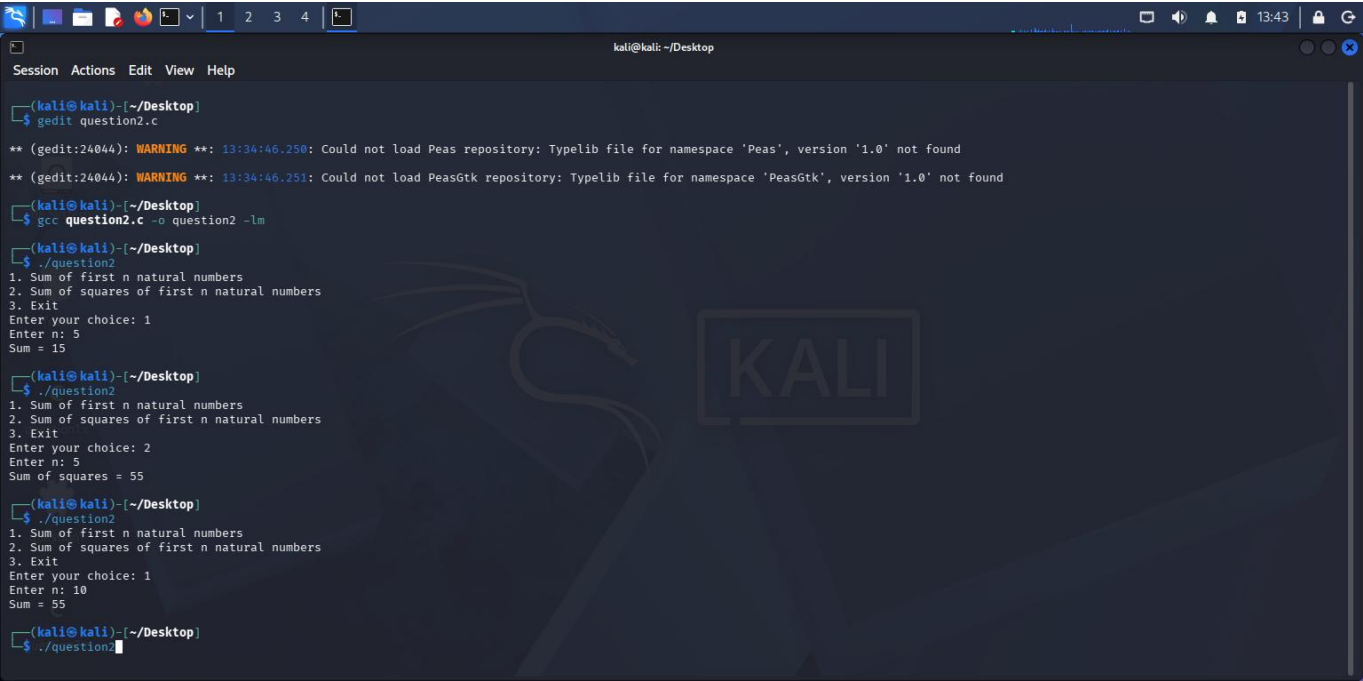
6. C Program

```
#include <stdio.h>
int main()
{
    int choice, n, i;
    int sum = 0;
    int sumSquares = 0;
    printf("1. Sum of first n natural numbers\n");
    printf("2. Sum of squares of first n natural numbers\n");
    printf("3. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    if (choice == 1)
    {
        printf("Enter n: ");
        scanf("%d", &n);
        for (i = 1; i <= n; i++)
        {
            sum = sum + i;
        }
        printf("Sum = %d\n", sum);
    }
    else if (choice == 2)
    {
        printf("Enter n: ");
        scanf("%d", &n);
        for (i = 1; i <= n; i++)
        {
            sumSquares = sumSquares + i * i;
        }
        printf("Sum of squares = %d\n", sumSquares);
    }
    else if (choice == 3)
    {
        printf("Exit");
    }
    else
    {
        printf("Invalid choice");
    }
    return 0;
}
```

7. Verification of All Test Cases

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	Choice 1; n = 5	Sum = 15	Sum = 15	Pass
2	Choice 2; n = 5	Sum of squares = 55	Sum of squares = 55	Pass
3	Choice 1; n = 10	Sum = 55	Sum = 55	Pass

8. Compilation and Execution Evidence



```
(kali@kali)-[~/Desktop]
└─$ gedit question2.c

** (gedit:24044): WARNING **: 13:34:46.250: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:24044): WARNING **: 13:34:46.251: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found

(kali@kali)-[~/Desktop]
└─$ gcc question2.c -o question2 -lm

(kali@kali)-[~/Desktop]
└─$ ./question2
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
3. Exit
Enter your choice: 1
Enter n: 5
Sum = 15

(kali@kali)-[~/Desktop]
└─$ ./question2
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
3. Exit
Enter your choice: 2
Enter n: 5
Sum of squares = 55

(kali@kali)-[~/Desktop]
└─$ ./question2
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
3. Exit
Enter your choice: 1
Enter n: 10
Sum = 55

(kali@kali)-[~/Desktop]
└─$ ./question2
```

QUESTION 3 – Interest Calculations

Write a menu-driven C program to calculate simple interest, compound amount, compound interest, and total payable amount for given principal, rate, and time.

1. Problem Understanding

- **Input:** Principal (P), rate of interest (R) in percent, and time (T) in years.
- **Output:** Simple interest, compound amount, compound interest, or total payable amount based on the selected menu option.
- **Formula:** $SI = (P \times R \times T) / 100$.
- **Formula:** $\text{Compound Amount} = P \times (1 + R/100)^T$.
- **Formula:** $\text{Compound Interest} = \text{Compound Amount} - P$.
- **Total Payable Amount:** For the compound-interest calculation, total payable amount is the compound amount.

Formula Summary

Quantity	Formula
Simple Interest	$SI = (P \times R \times T) / 100$
Compound Amount	$A = P \times (1 + R/100)^T$
Compound Interest	$CI = A - P$

Total Payable Amount	Total = A
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2. Standard Test Cases

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	Choice = 1, P = 1000, R = 10%, T = 2 years	SI = 200.00	SI = 200.00	Pass
2	Choice = 2, P = 1000, R = 10%, T = 2 years	Compound Amount = 1210.00	Compound Amount = 1210.00	Pass
3	Choice = 3, P = 1000, R = 10%, T = 2 years	CI = 210.00; Total = 1210.00	CI = 210.00; Total = 1210.00	Pass

3. Algorithm

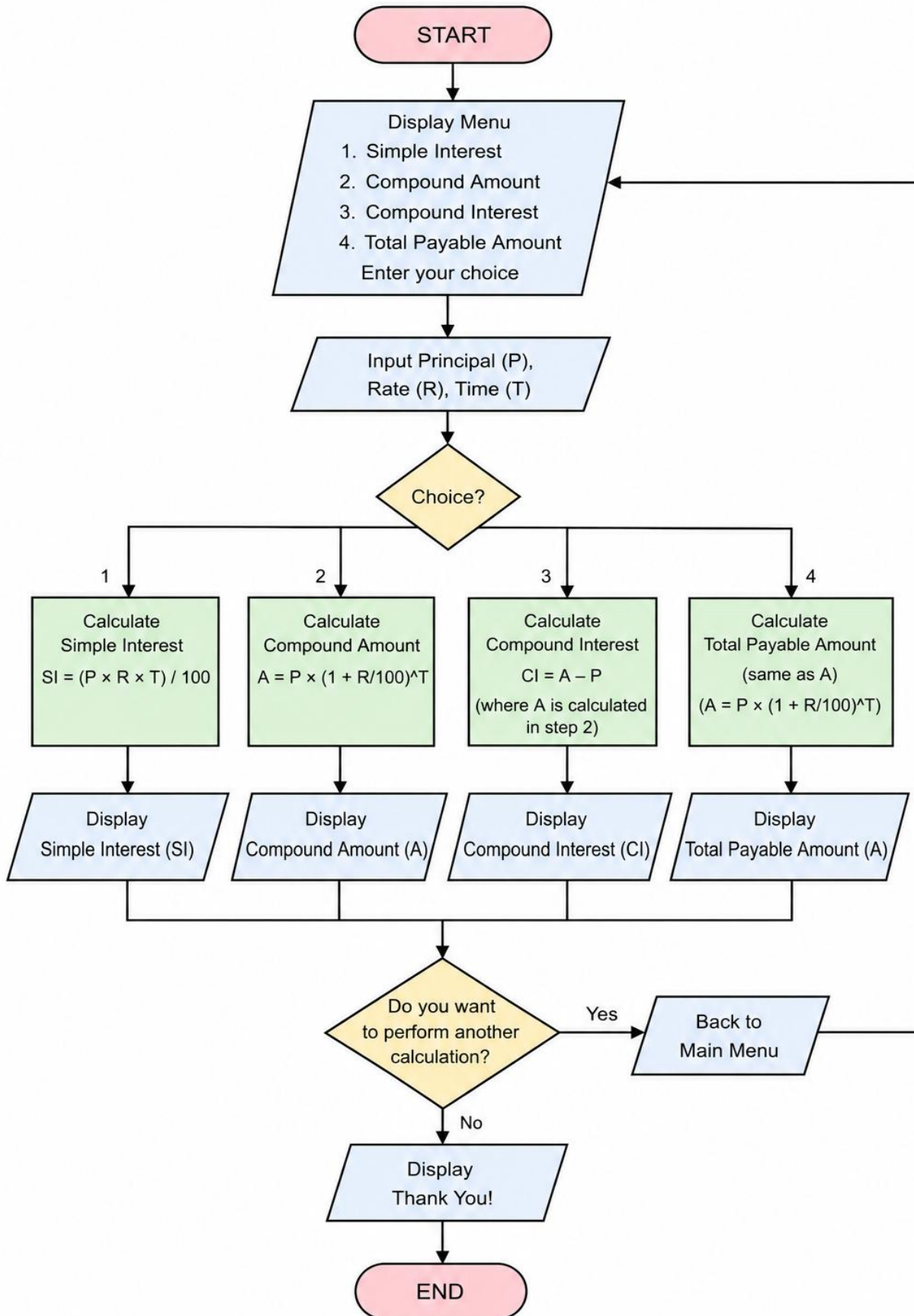
1. Start the program.
2. Declare variables for choice, principal, rate, time, simple interest, compound amount, and compound interest.
3. Display the interest menu and read the choice.
4. If choice is from 1 to 4, read principal, rate, and time.
5. Calculate simple interest.
6. Calculate compound amount using the power function.
7. Calculate compound interest as compound amount minus principal.
8. Use the selected menu option to display the required value.
9. If choice is not 1 to 4 or 5, display an invalid-choice message.
10. Repeat until Exit is selected.
11. Stop the program.

4. Pseudocode

1. BEGIN
2. REPEAT
3. Display interest menu.
4. Read CHOICE.
5. IF CHOICE is 1, 2, 3, or 4 THEN
6. Read P, R, T.
7. $SI \leftarrow (P \times R \times T) / 100$.

8. $A \leftarrow P \times (1 + R/100)^T$.
9. $CI \leftarrow A - P$.
10. IF CHOICE = 1 display SI.
11. ELSE IF CHOICE = 2 display A.
12. ELSE IF CHOICE = 3 display CI.
13. ELSE display A as Total Payable Amount.
14. ELSE IF CHOICE $\neq$ 5 THEN display Invalid Choice.
15. END IF.
16. UNTIL CHOICE = 5.
17. END

Flowchart – Question 3



6. C Program

```
#include <stdio.h>
#include <math.h>
int main()
{
    int choice;
    float p, r, t;
    float si, amount, ci;
    printf("1. Simple Interest\n");
    printf("2. Compound Amount\n");
    printf("3. Compound Interest\n");
    printf("4. Total Payable Amount\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    printf("Enter principal: ");
    scanf("%f", &p);
    printf("Enter rate: ");
    scanf("%f", &r);
    printf("Enter time: ");
    scanf("%f", &t);
    si = (p * r * t) / 100;
    amount = p * pow((1 + r / 100), t);
    ci = amount - p;
    if (choice == 1)
    {
        printf("Simple Interest = %.2f", si);
    }
    else if (choice == 2)
    {
        printf("Compound Amount = %.2f", amount);
    }
    else if (choice == 3)
    {
        printf("Compound Interest = %.2f", ci);
    }
    else if (choice == 4)
    {
        printf("Total Payable Amount = %.2f", amount);
    }
    else
    {
        printf("Invalid choice");
    }
    return 0;
}
```

7. Verification of All Test Cases

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
-----------	-------	-----------------	---------------	--------------------

1	Choice 1; P = 1000, R = 10%, T = 2	SI = 200.00	SI = 200.00	Pass
2	Choice 2; P = 1000, R = 10%, T = 2	Compound Amount = 1210.00	Compound Amount = 1210.00	Pass
3	Choice 3; P = 1000, R = 10%, T = 2	CI = 210.00	CI = 210.00	Pass
4	Choice 4; P = 1000, R = 10%, T = 2	Total Payable = 1210.00	Total Payable = 1210.00	Pass

8. Compilation and Execution Evidence

```

kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)-[~/Desktop]
└─$ gcc question3.c -o question3 -lm

(kali@kali)-[~/Desktop]
└─$ ./question3
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
Enter your choice: 1
Enter principal: 1000
Enter rate: 10
Enter time: 2
Simple Interest = 200.00

(kali@kali)-[~/Desktop]
└─$ ./question3
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
Enter your choice: 2
Enter principal: 1000
Enter rate: 10
Enter time: 2
Compound Amount = 1210.00

(kali@kali)-[~/Desktop]
└─$ ./question3
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
Enter your choice: 3
Enter principal: 1000
Enter rate: 10
Enter time: 2
Compound Interest = 210.00

(kali@kali)-[~/Desktop]
└─$

```